

ACFI315 Principles of finance with Excel

Week 3 Forecasting in Excel (Polynomial Relationship)

A number of data files have been uploaded to VITAL.

1. Next Sales Data

Research questions:

- (a) Calculate the growth revenue for Next. Use the growth from 2018-2019 to forecast revenue for the next two financial years.
- (b) Fit a linear time trend to Next's revenue. Use the result to forecast revenue for the next two years.
- (c) Fit a second order polynomial to the revenue data. Extract the coefficients using the LINEST function and forecast revenue for the next two years.
- (d) Fit a third order polynomial to the revenue data. Extract the coefficients using the LINEST function and forecast revenue for the next two years.
- (e) Next, like many retailers, helpfully provides available square feet of sales space. Fit a linear regression of revenue (y-variable) floor space (x-variable) and use the result to forecast revenue for the next financial year. In the 2019 annual report Next state that they "we expect that trading space will increase by around 50,000 square feet during 2019, subject to lease renewal negotiations. We expect new stores to add +160k sq ft, mainline store closures to deduct -60k sq ft and clearance store closures to deduct -50k sq ft." use this to forecast for the next year.
- (f) Use Data – Data Analysis – Regression to run an ordinary least squares (OLS) regression of Revenue (y-variable) on two x-variables, square foot and year. Use the coefficients and the new X-values to forecast one period ahead.

2. Apple Sales Data from Week 2

Research questions:

- (a) Create a data table for Apple's 2019 Total Operating Income assuming Sales Growth Varies from -5% to +15% in 2.5% in increments AND R&D as a % of sales varies from 2% to 10% in increments of 2%.
- (b) Create a data table for Apple's 2019 Total Operating Income assuming Sales Growth Varies from -5% to +15% in 2.5% in increments AND S,G&A as a % of sales varies from 5% to 13% in increments of 2%.